NOTHING PHONE (3A) PRO · A059P

StarStacker Field Guide

Every screen, every number, and the order to press things in. Built for reading on the phone in the dark — it stays on the red end of the palette so it will not cost you your dark adaptation.

IF YOU READ ONE THING

Framing and focus happen before you press Start. Once a session is running there is no picture on screen — only numbers. So get the framing right, get focus locked, and only then start.

This is the app's first night out. The exposure engine, the focus sweep and the pointing maths have never been run against real stars. Expect to find things. Every frame is kept on disk regardless, so a session is never wasted.

The five screens

1 · Probe	What the phone can do. Where you resume an interrupted session.
2 · Framing	See the sky, point it, lock focus.
3 · Setup	Measure the sky, solve the exposure, check the plan.
4 · Capturing	Numbers only. Pause, or end and take darks.

5 · Complete

What you got, and where it is.

DO THIS

Tonight, in order

1. **Set up the tripod and let the phone sit.**

   Give it a minute to stop wobbling before you focus. A tripod still settling will spoil a focus sweep.

2. Open StarStacker. Scroll down. Tap **Framing & focus**

   Camera 0 · Main is already selected and is the right one — 2.00 µm pixels at f/1.88 beats every other lens on this phone.

3. Tap **Start night preview**

   First frame takes ~3 s, then one per second. The line under the image says the rate so a slow preview does not read as a frozen app.

4. **Point it.** Watch the preview and the `Stars` count.

   If it is too faint, tap Boost · 4 s. Put it back to 1 s before focusing.

5. Check the **Pointing** card shows a number for `Declination at centre`.

   If it says needs location, you have no GPS fix. Not fatal — the app assumes the celestial equator, which just gives you shorter subs than you need. Opening Maps for a moment usually fixes it.

6. Tap Sweep focus and **do not touch the phone** for ~1.5 minutes.

   Nine lens positions. Look for `LOCKED` and a clear dip in the bar chart. If it fails, see *When focus will not lock*.

7. Tap Continue to session setup →

8 Tap **Measure the sky** — about a minute.

It walks nine ISOs reading the sensor's noise, then measures how bright your sky actually is.

9 **Read the one line.** Something like `ISO 800 · 12 s · 150 frames · 30 min`.

Tap **Show work** if you want to see why it chose that, and every ISO it rejected. Tap any ISO to pin it.

10 Pick a length: **15m** **30m** **60m** **120m**

Check **`Storage`** and **`Battery`** are not amber. Start is disabled if the session genuinely will not fit.

11 Tap **Start session**

12 **Press the power button** to blank the screen and walk away.

Capture keeps running — it lives in a foreground service with a wake lock, not in the app window.

13 When the lights finish, the phone shows and notifies **Cover the lens**.

Cap it or press it face-down on something opaque, then **Lens is covered — take darks**. No answer within 15 minutes and it finishes without darks and records why.

14 Read the completion screen. **Note the folder path.**

The preview stops itself after 2 minutes of no interaction. That is deliberate — framing at ISO 3200 warms the sensor, and that heat comes out of the session you have not shot yet. Tap anything to keep it alive.

REFERENCE

Every number on screen

The four live metrics

HFR px	**Half-flux radius** — the radius holding half the star's light. **The focus number. Lower is sharper.** Measured on the binned 1024×768 analysis image, so one HFR pixel is about four sensor pixels. Judge it by minimising it, not against an absolute.
Stars	How many stars were detected in this frame. **A relative number, not a score** — a wide lens on the Milky Way gives hundreds, a narrow field gives dozens. What matters is a sudden *drop*, which means cloud.
Ecc	**Eccentricity.** 0.00 is a perfect circle, 1.00 is a line. Round stars sit around 0.2–0.4 in practice. Above 0.60 the frame is flagged Trailed — the sub is too long, or something moved.
Sky ADU	Median background brightness in raw sensor counts. On this sensor **64 = black** (no light at all) and **1023 = clipped**. 65–80 is a genuinely dark sky. 300+ is heavy light pollution. 1023 means the frame is ruined and nothing can be measured from it.

Units you will see

`ADU`	Analogue-to-Digital Unit — the raw count out of the sensor's converter. 10-bit here, so 0–1023.
`e⁻`	Electrons. The physical unit of collected light, and of read noise. Used because you cannot compare sky brightness to sensor noise in ADU — only in electrons.
`e⁻/s`	How fast the sky fills a pixel. **Independent of ISO** — ISO is gain applied after the light is collected, so the app measures this once.
`stops`	Doublings. "1.6 stops of headroom" means the sky could get three times brighter before it clips.
`dioptres`	Focus distance as 1/metres. **0 = infinity**, larger = closer. This lens moves in steps of 0.0374. Asking for exactly 0.0 gets you 0.1216 (the hyperfocal position, 8.2 m) — a quirk of this phone.
`″/s`	Arcseconds per second. An arcsecond is 1/3600 of a degree.

REFERENCE

Numbers in the solve

`Trailing limit`	The longest single exposure before stars smear more than you allowed. Computed from your *measured* pixel pitch and focal length and the declination you are pointing at — not the "500 rule".
`Star elongation budget`	How much smear you will accept, in pixels. Default 1.5.
`Sky background`	Your sky's brightness in e^-/s, measured from a real test frame rather than guessed from a light-pollution map.
`Read noise`	Noise added by reading the sensor, in electrons. On this phone: 5.6 e^- at ISO 50 falling to about 2.0 e^- at ISO 3200 and flat above that.
`Dual-gain switch`	The ISO where read noise drops off a cliff on sensors that have one. **This sensor has none** — it declines smoothly.
`Highlight headroom`	Stops between the sky level and clipping. More headroom = fewer blown-out stars.
`Sky vs read noise`	The whole point of the engine. Once the sky's own shot noise is **3–5× the sensor's read noise in variance**, a

longer sub buys you almost nothing and costs you trailing. That is "sky-limited".

The two answers it can give

"sky-limited …" — good. The sub is long enough that your sky, not the sensor, sets the noise. If it adds "*set by clipping*", your sky is bright and highlight headroom is what is limiting you.

"read-noise limited …" — also fine, and it means **you have a dark sky.** No ISO can reach sky-limited inside the trailing limit. The app shoots at the trailing limit and the answer is to stack more frames, not longer ones.

In the plan

`Frame left at end`	How much of the frame survives field rotation over the session. On a fixed tripod the sky turns as well as drifts, so the edges are not shared by every sub. Below 80% it turns amber.
`Field rotation`	How fast the frame turns, arcsec/s. Fastest looking south, and it diverges near the zenith. This does not blur one sub — it decides how much you keep.
`Declination`	Angular distance from the celestial equator. Stars near the pole barely move; stars on the equator move fastest. This sets the trailing limit.

`Storage`	25 MB per frame, measured. Amber near the limit, and Start is blocked if it will not fit.
`Battery`	An estimate, not a measurement. Tonight is what produces the real number.

SCREEN 4

While it is shooting

No picture — numbers only. The big ring is frames done out of frames planned; you can read it from across a field without reading anything.

The frame line

`HFR` · **`Stars`** · **`Sky`** · **`Kept`**, then a smaller line with **`headroom`**, battery temperature and charge.

`Kept`	Frames accepted so far. Turns amber if more than half are being rejected.
`headroom`	Thermal headroom. 0 = cool, 1.0 = at the throttling point. Pacing is switched off deliberately for now — it only logs.

Why frames get rejected

`Trailed`	Eccentricity over 0.60 — stars are streaks. **Shorten the sub.**
`Cloud`	Star count fell below half the running baseline. **Wait.**
`Bumped`	The phone tilted more than 1.0° during the exposure. **Steady the tripod.**
`Saturated`	The frame is clipped — a car, the Moon, a torch. **Too much light.**

Nothing is ever deleted. Rejected frames are written to `lights/` like any other and flagged in `session.json` with the number that flagged them. You can include them later if you disagree.

Controls

Pause stops between frames and can be resumed. **End & take darks** skips the rest of the lights and goes straight to the darks prompt.

What you get at the end

```
Android/data/com.starstacker/files/sessions/<date>_<label>/
```

with `lights/`, `darks/`, `flats/`, `master/` and `session.json`. The DNGs open in Siril, DeepSkyStacker or RawTherapee as they are — reachable over USB without the app's help.

WHEN IT GOES WRONG

When focus will not lock

You are never blocked. With no stored focus the app shoots at the hyperfocal position, which has infinity inside its depth of field. A session with no focus step is *soft, not ruined*. Do not abandon a clear night over this.

Read the verdict

`TOO_FEW_STARS`	The commonest one. It needs 5+ stars at 3 of the 9 positions. **Check the live Stars count first** — under ~20 and the sweep cannot work. Try Boost · 4 s (slower, ~6 min), get away from horizon glow, or use the trick below.
`FLAT`	HFR barely changes across the sweep. Usually means you are **measuring hot pixels, not stars** — they do not change shape with focus. A few hundred "stars" in an empty-looking sky is the giveaway.
`MINIMUM_AT_EDGE` far end	**Not a failure.** "The furthest this lens reached" is the expected answer for stars. It stores it and reads LOCKED. Carry on.

THE TRICK WORTH KNOWING

Focus is stored per camera, not per target. Point at a dense star field, sweep there, then re-point at whatever you actually want to shoot. The focus follows you.

Focus by hand

Under **FOCUS BY HAND** on the framing screen:

1. ◀ Further / Nearer ▶ — one tap is one real motor step.
   Allow ~2 s per tap for the number to catch up. The sensor pipeline here is ten frames deep.
2. Watch **HFR** on that row. **Lower is sharper.**
   Star count usually rises as you approach focus, as faint stars climb over the detection threshold.
3. Walk it down, overshoot by one, come back.
4. Tap Use this focus for the session

WHEN IT GOES WRONG

Everything else

Session died	Reopen the app. The **Unfinished session** card is on the first screen with Resume. It carries on filling the same folder from the next frame, reusing the exposure and focus it already decided. Leave it only stops the offer — every frame stays on disk.
"Measure the sky" errors immediately	The camera is probably still closing from the framing loop. Wait two seconds and tap it again.
Everything says Saturated	Something is lighting the frame — a streetlight, the Moon just out of frame, a phone screen. Re-point, or drop the framing ISO.
Preview is a black rectangle	Try Boost · 4 s. If still black, the lens is likely covered or pointing at ground.
Compass accuracy is low	Wave the phone in a figure of eight. Azimuth feeds declination, which sets the trailing limit.
It stopped overnight	Battery optimisation is the first suspect — the app is not whitelisted. Worth noting what time it stopped.

What to bring back

Whatever happens, `session.json` in the session folder records every frame with its ISO, exposure, temperature, HFR, star count, background and the exact reason for any rejection. **That file answers almost any question about what happened** — it beats trying to remember.

Known-untested tonight

- Whether the night preview actually shows stars
- Whether the focus sweep finds a real minimum, and whether two sweeps agree
- Whether alt/az matches a known star
- Whether the solved sub length is sensible under a real dark sky

Those four are exactly what tonight exists to answer. Everything else — capture, writing, gating, darks, resume — has been demonstrated on this phone.